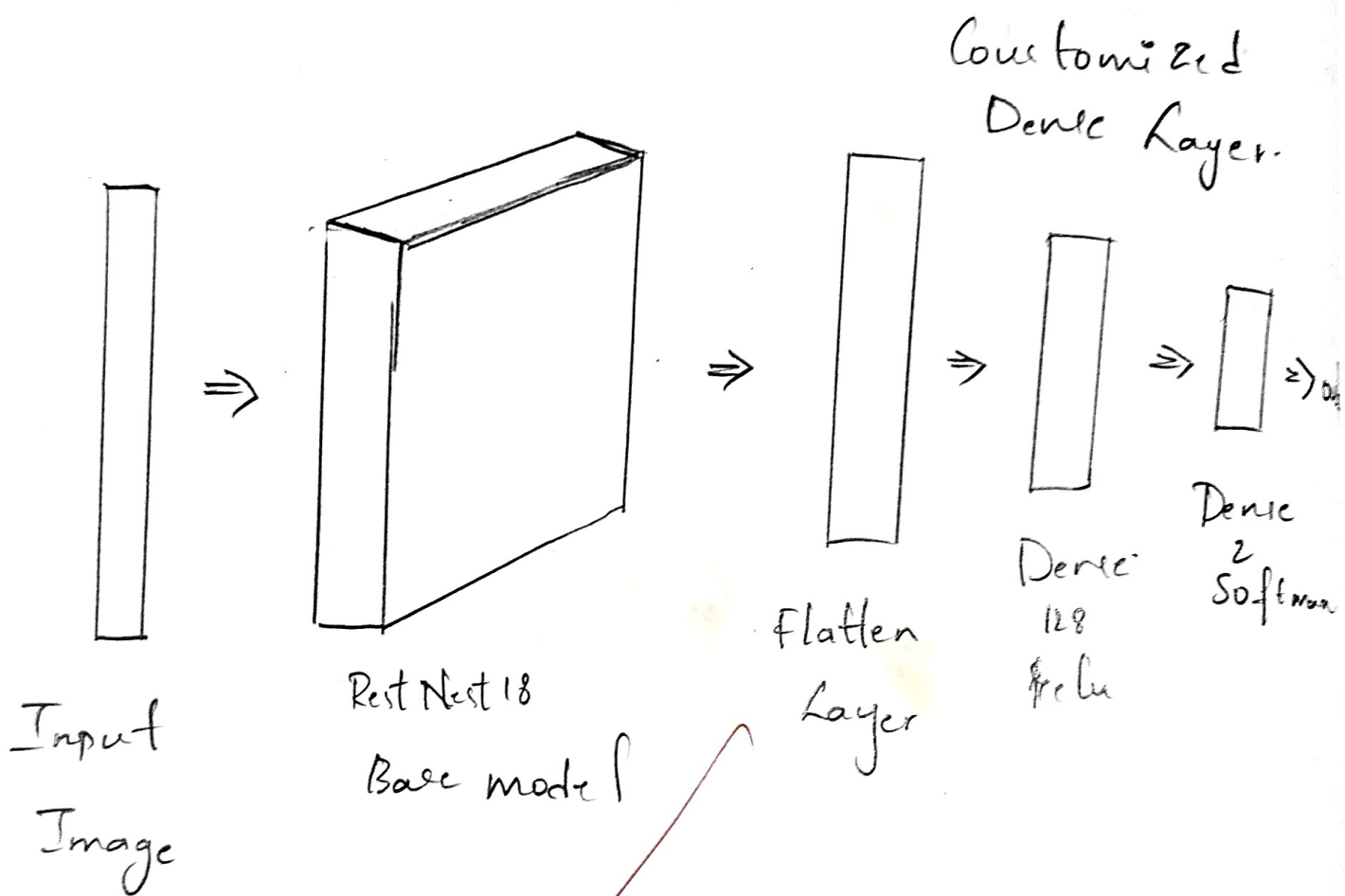


Architecture:



Exp:-14

Implement a Pre Trained CNN Model
on a feature extraction Using transfer learning

Aim: To Perform image classification on CIFAR-10 dataset Using Transfer learning with a Pre trained ResNet 18 model

Objective:-

- To Understand and implement transfer learning Using a Pre trained ResNet 18 model
- To train the modified model and Visualize the Loss Curve

Pseudo Code :-

- Begin
- Import torch, torchvision, torch.nn
- Set device
- Load Pre-trained ResNet 18 model
- Freeze all layers.
For each parameter in model
Set requires_grad = False
- Modify the final layer for CIFAR-10:
train_loader ← CIFAR10 (train = True)
- Define transformation:
transform ← Resize (224x224) + ToTensor;

Observation :

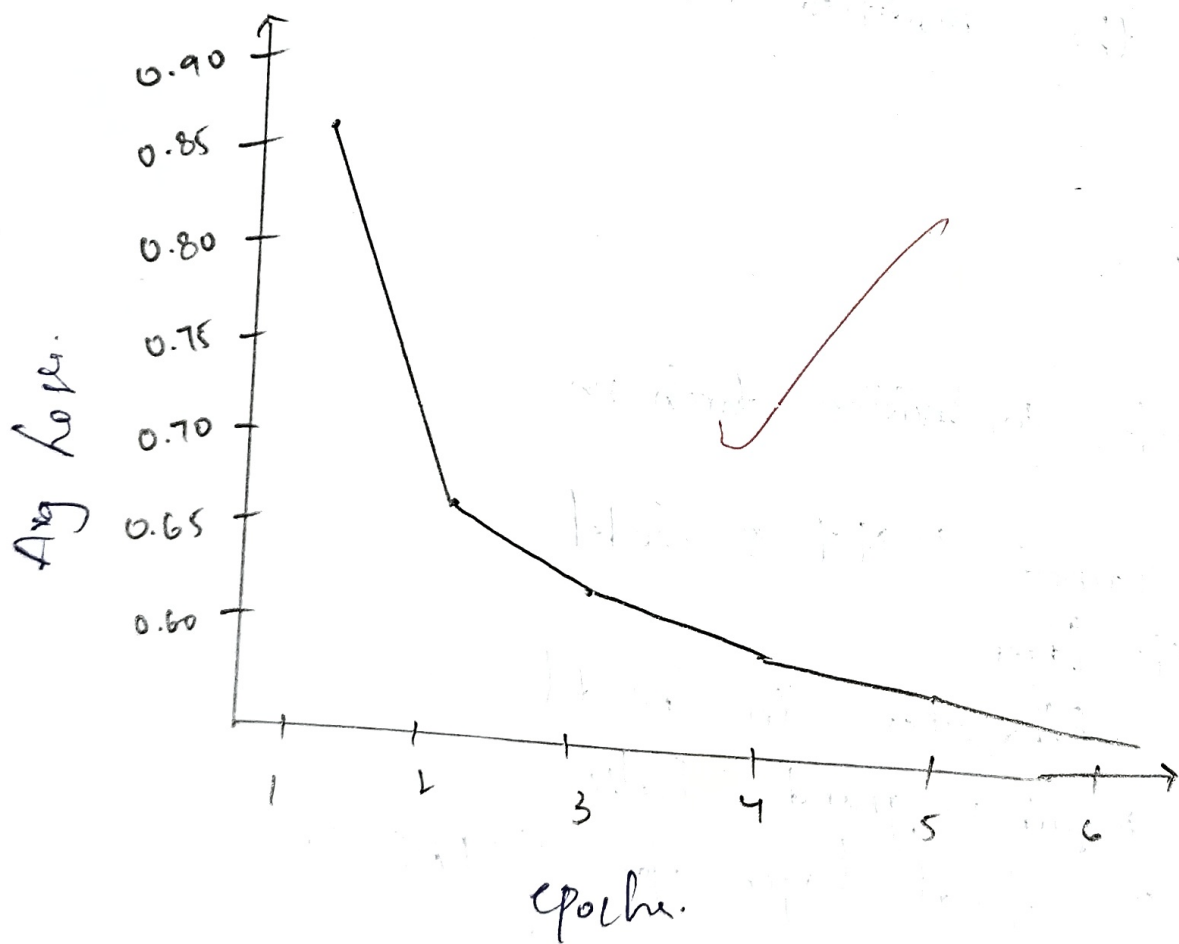
Epoch 1 : loss : 0.8348

Epoch 2 : loss : 0.6289

Epoch 3 : loss : 0.5937

Epoch 4 : loss : 0.5997

Epoch 5 : loss : 0.5699



- 11/1/2023
- Define loss function and optimizer:
criterion $\leftarrow$ CategoricalCrossentropyLoss()
 - For each epoch in range(1 to 10):
 - avg loss $\leftarrow$ running loss / total-batches
 - append avg-loss to loss-values
 - Print epoch number and avg loss
 - Print "Training Complete"
 - Plot loss Curve:
 - x-axis $\leftarrow$ epochs
 - y-axis $\leftarrow$ average loss
 - End:

Result: Successfully implemented a Pretrained CNN model.

~~21/1~~

Total 4/1/2023